Using Time Series Analysis for Sales and Demand Forecasting

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1 Introduction

The ISBN data consists of 483 unique books, while the weekly sales data has 227224 original rows. This project uses these data as sales data over a perioed of about 24 years.

1.1 Our Approach

We have followed the list of activities given in the project's rubric. The ISBN data set contained the unique records of 500 books, while the sales record was provided in the second Excel file. In fact, there are multiple ISBNs corresponding to a single Title. The Google colab could not load MS Excel files for the two data sets successfully and generated a BadZipFile error. We tried a few work-arounds but internet research reported that there was a Microsoft sensitivity tag that prevented the loading of the file with multiple worksheets. That tag can only be removed by the original creator of this file. So we resorted to save every worksheet as a .CSV UTF-8 file separately on the local hard disk and then loaded them as Pandas dataframes, which Pandas can concatenate into one dataframe. This was repeated for the second data set too.

2 Initial Data Investigation

This included obtaining data for two specific books *The Alchemist*, hereinafter referred to as Book1, and *The Very Hungry Caterpillar*, hereinafter referred to as Book2.

W e have selected the Volume column to analyse and forecast sales data. The ISBNs were converted into string type as required, and data was resampled on weekly basis. Dates were converted into datetime object. A specific requirement was to collect ISBNs of those books whose sales data existed beyond the date July, 01, 2024. The query returned with 308 ISBNs. These results that are depicted in Figure 2. The sales data for these ISBNs has been depicted in Figure 1.

S ales patterns for the periods from January 1, 2001 to December 31, 2012, and for the period January 01, 2013 to December 31, 2024 for all books are depicted in Figures 3 and 4 respectively.

A s required, we filtered the sales of the above-mentioned two books, called Book 1 and Book 2, these are shown in Figures 5 and 6.

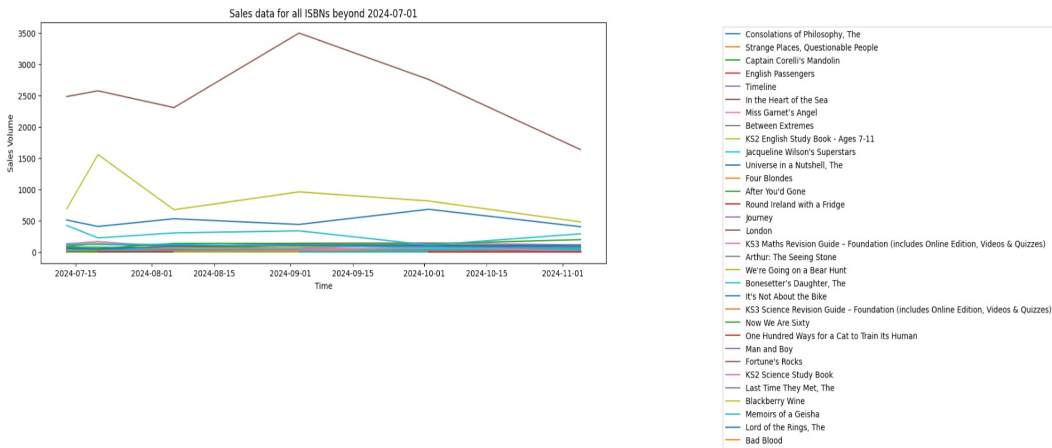


Figure 1: Plot of sales data for ISBNs from July 01, 2024 onwards.

```
print(beyond01072024df['ISBN'].unique())
```

```
['9780140276619' '9780330355667' '9780749397548' '9780140285215'
 '9780099244721' '9780006531203' '9780006514213' '9780552145954'
 '9781841461502' '9780440864554' '9780593048153' '9780349114033'
 '9780747268161' '9780091867775' '9780552145060' '9780099422587'
 '9781841460406' '9780752844299' '9780744523232' '9780006550433'
 '9780224060875' '9781841462400' '9780719559792' '9780340786055'
 '9780006512134' '9780349112763' '9781841462509' '9780349113609'
 '9780552998000' '9780099771517' '9780261103252' '9781841150437'
 '9781841462301' '9780440864141' '9780140295962' '9780006514091'
 '9780140275421' '9780552997348' '9780006647553' '9780091816971'
 '9781841460307' '9780007101887' '9780140281293' '9780340766057'
 '9780752846576' '9780241003008' '9780552998482' '9780099285823'
 '9780099286578' '9780140276336' '9780722532935' '9780340696767'
 '9780140294231' '9780552145053' '9780552998444' '9780099428558'
 '9780099286387' '9780552998727' '9780140259506' '9780091816650'
 '9780552996457' '9780552145817' '9780552996006' '9780552141352'
 '9780006510901' '9780006531777']
```

Figure 2: ISBNs for the books with sales data present from July 01, 2024 onwards.

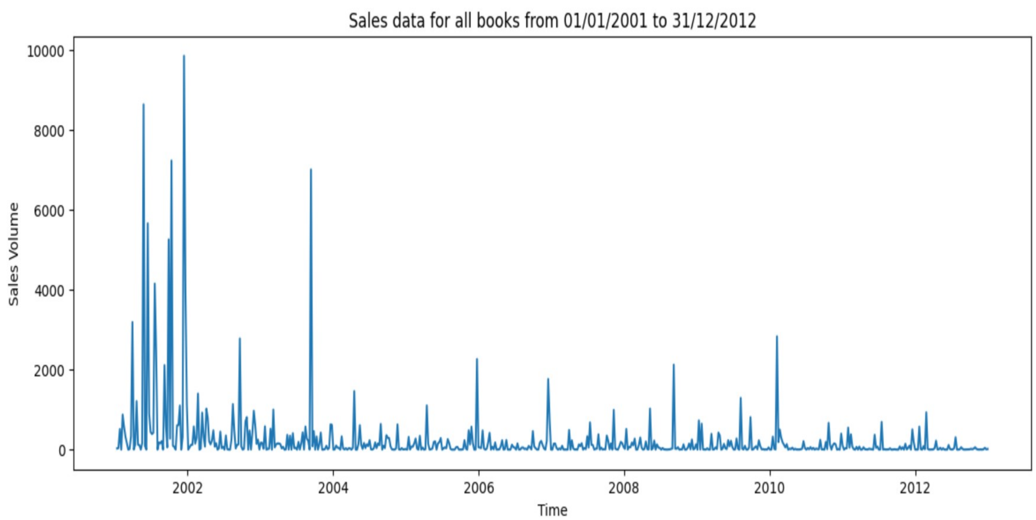


Figure 3: Sales of all books from July 01, 2001 to December 31, 2012.

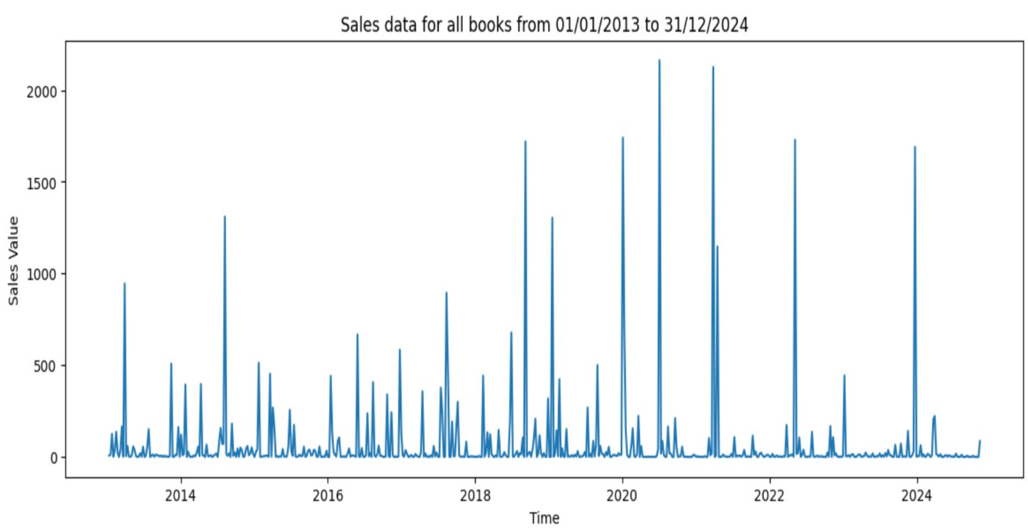


Figure 4: Sales of all books from July 01, 2013 to December 31, 2024.

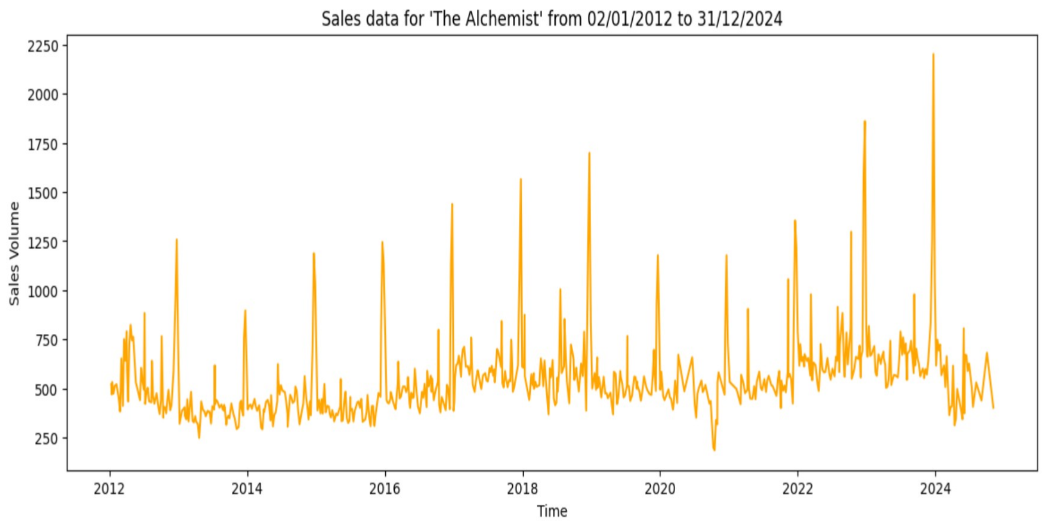


Figure 5: Sales data for The Alchemist from Janiary 01, 2001 onwards.

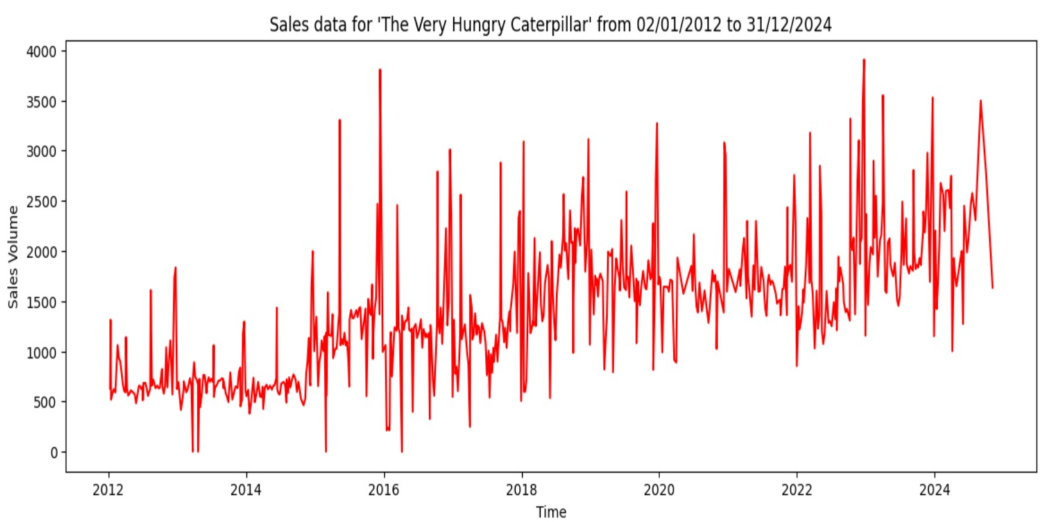


Figure 6: Sales data for The Very Hungry Caterpillar from Janiary 01, 2012 onwards.

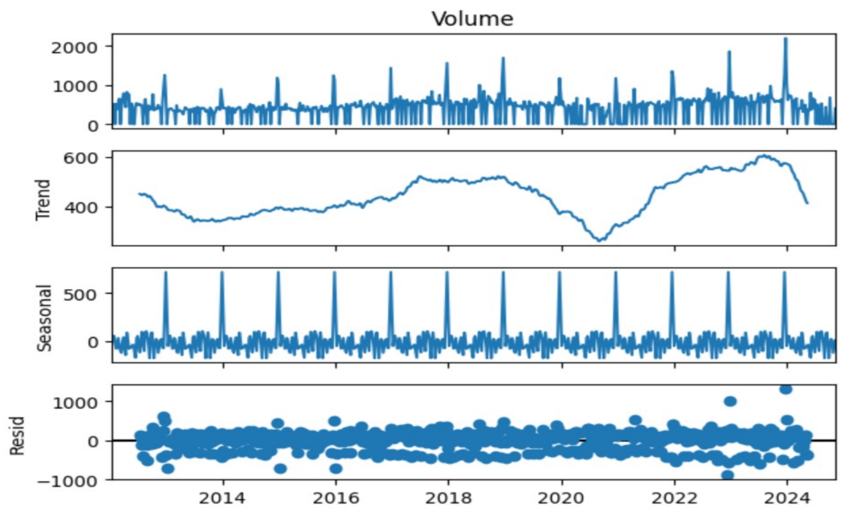


Figure 7: Seasonal docomposition of The Alchemist sales data from January 01, 2012 onwards.

3 Classical Techniques

3.1 Decomposition, ACF and PACF Plots & Stationarity

3.1.1 Book 1: The Alchemist

Seasonal decomposition, the ACF and PACF plots of `Book1` are depicted in Figures 7, 8a and 8b respectively.

Stationarity: The p-value for `Book1` from the ADF test were as follows:

p-value from original data: 0.01856723378220044
p-value from differenced data: $5.497638145593545e - 20$

We notice that the very small p-value for the original data which means that we reject the null hypothesis that there is insuffieint evidence to establish that the `Book1` data has no unit root and thus there is stationarity in Book 1 sales data. However, when we apply differencing, the p-value from ADF method is very small and indicates that the data has no unit root and is stationry.

3.1.2 Book 2: The Very Hungry Caterpillar

Seasonal decomposition, the ACF and PACF plots of `Book2` are depicted in Figures 9, 10a and 10b respectively.

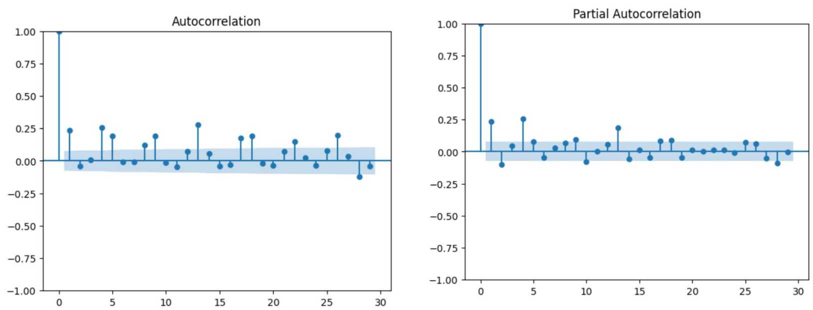


Figure 8: The Alchemist sales data from January 01, 2012 onwards: (a) The ACF plot, (b) The PACF plot

Stationarity: The p-value for Book2 from the ADF test were as follows:

p-value. original data: $3.5612961122124380e-25$
p-value. differenced data: $3.5612961122124380e-25$

For Book2, the p-value of original data is also very small which means that we can reject the hypothesis that data has a unit root, so it does have a staionarity. The p-value for the difference data is also very small causing the rejection of the null hypothesis.

3.2 Classical Approaches applied at The Alchemist Sales Data

3.2.1 The SARIMA and Auto ARIMA Models

Table 1 lists all of the SARIMA model parameters attempted for Book1 before running Auto ARIMA. It also lists down the performance results for these models. From the Table, it is clear that the SARIMA5 model produced the lowest MSE. Hence, we discussed the residuals by this model.

3.2.2 Comments on Residuals

Figure 11 depicts the residuals by SARIMA5 models for Book1. Figure shows residuals fluctuating around the mean 0 but with slight change in variance. The density function shows a near normal distribution curve with a small hint of left skew. This is shown through the deviation of a couple of residuals away from the red line. The ACF plot shows some autocorrelation.

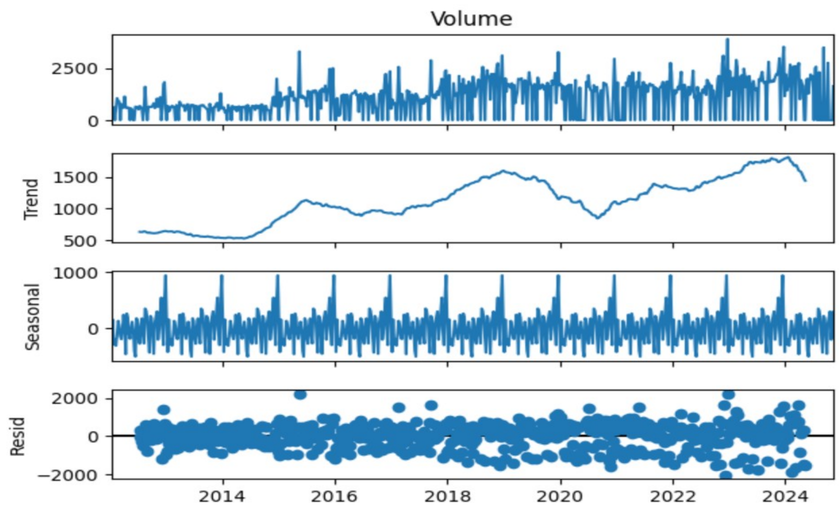
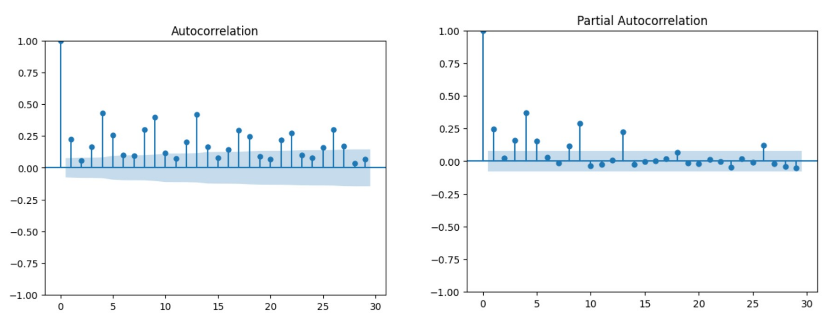


Figure 9: Seasonal docomposition of The Very Hungry Caterpillar sales data from January 01, 2012 onwards.



(a)

(b)

Figure 10: The Very Hungry Caterpillar sales data from January 01, 2012 onwards: (a) The ACF plot, (b) The PACF plot

Model	Parameters: (p, d, q), (P, D, Q, m)	AIC Score	Mean squared error (MSE)
SARIMA1	$(1, 0, 0), (1, 1, 0, 52)$	8246.925	221682.2710791234
SARIMA2	$(0, 0, 1), (0, 1, 0, 52)$	8361.046	219012.63145667798
SARIMA3	$(0, 0, 0), (0, 1, 1, 52)$	8201.802	170962.11634679642
SARIMA4	$(0, 1, 1), (1, 1, 0, 52)$	8201.494	255768.67350706225
SARIMA5	$(1, 0, 1)(1, 1, 1, 52)$	8825.905	169842.38895730104
Auto ARIMA SARIMAX	$(2, 1, 2), (1, 0, 1, 52)$ with ranges $0 \leq p \leq 2, 0 \leq q \leq 2, d = 1$ and $0 \leq P \leq 1, 0 \leq Q \leq 1, D = 1, m = 52$	8825.834	197392.34054717518
Auto ARIMA SARIMAX	$(3, 1, 1), (1, 0, 1, 52)$ with ranges $0 \leq p \leq 3, 0 \leq q \leq 3, d = 0$ and $0 \leq P \leq 1, 0 \leq Q \leq 1, D = 1, m = 52$	8819.054	191243.91024723492

Table 1: SARIMA and Auto ARIMA model parameters and results for The Alchemist sales data from January 01, 2012 onwards.

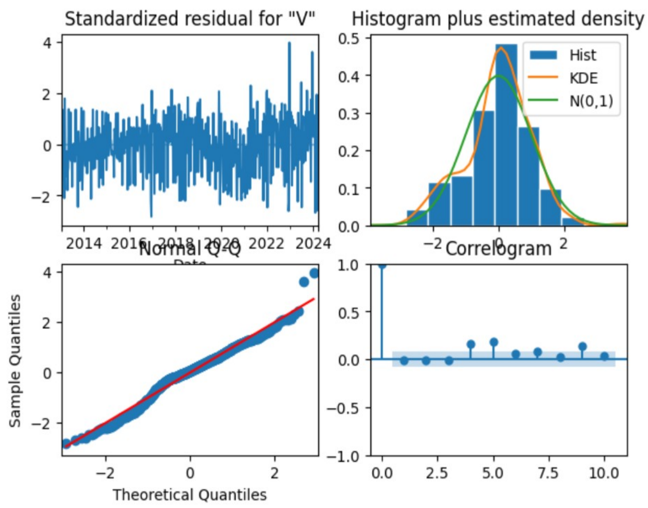


Figure 11: Residuals for SARIMA5 model applied to The Alchemist sales data from January 01, 2012 onwards.

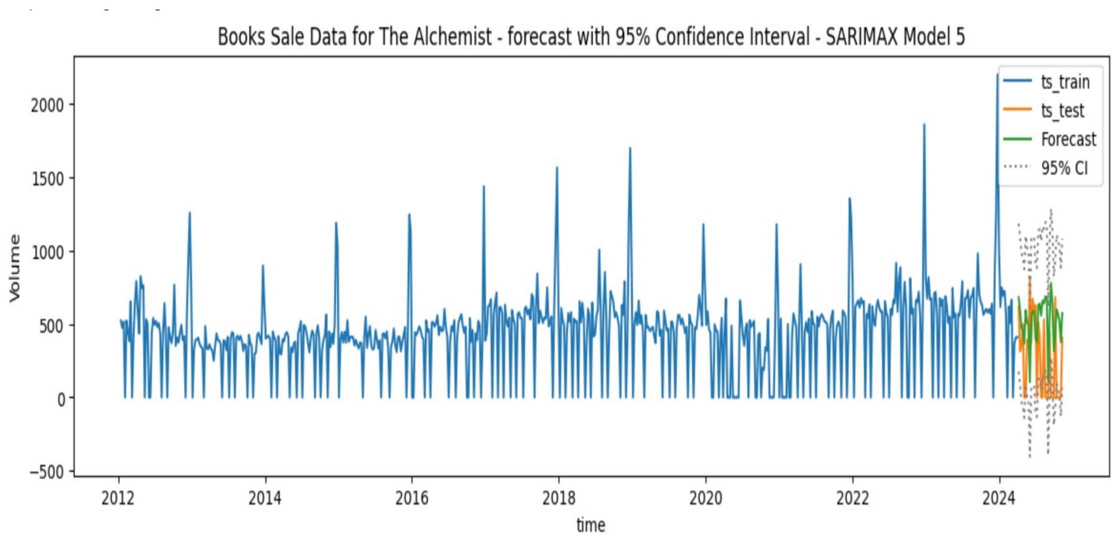


Figure 12: 32-Weeks prediction using SARIMA5 model for The Alchemist using sales data from January 1, 2012.

3.2.3 Prediction for the Final 32 Weeks

Predictions for the final 32 weeks of sales for the Book1 are depicted in Figure 12. These predictions fit well with test data for this model.

3.3 Classical Approaches applied at The Very Hungry Caterpillar Sales Data

3.3.1 The SARIMA and Auto ARIMA Models

We attempted six SARIMA models before attempting the Auto ARIMA model. These are detailed in Table 2. The results show that SARIMA3 was the best model having the least MSE and AIC scores.

3.3.2 Comments on Residuals

Figure 13 depicts the residuals for the SARIMA3 model. The residual errors seem to fluctuate around a mean of zero and have a changing variance for the Book2 sales data. The density plot suggests a good normal distribution. All the dots should fall perfectly in line with the red line. Any significant deviations would imply the distribution is skewed. We can see that some points deviate from the red line. The ACF plot shows the residual errors are autocorrelated. Any autocorrelation would imply that there is some pattern in the residual errors.

Model Name	Parameters: (p, d, q), (P, D, Q, m)	AIC Score	Mean squared error (MSE)
SARIMA1	$(1, 0, 0), (1, 1, 0, 52)$	9445.662	1805853.689730868
SARIMA2	$(0, 0, 1), (0, 1, 0, 52)$	9578.990	2090220.5593585717
SARIMA3	$(0, 0, 0), (0, 1, 1, 52)$	9391.188	1666209.4373392833
SARIMA4	$(0, 1, 1), (1, 1, 0, 52)$	9398.201	2393945.937094783
SARIMA5	$(1, 0, 1)(1, 1, 1, 52)$	9391.086	1710626.7063489635
SARIMA6	$(2, 0, 2)(1, 1, 1, 52)$	9391.086	1954277.0429303243
Auto ARIMA SARIMAX	$(0, 1, 1), (1, 0, 1, 52)$ with ranges $0 \leq p \leq 2, 0 \leq q \leq 2, d = 0$ and $0 \leq P \leq 1, 0 \leq Q \leq 1, D = 1, m = 52$	10092.856	2041512.0289632597
Auto ARIMA SARIMAX	$(0, 1, 1), (2, 0, 0, 52)$ with ranges $0 \leq p \leq 3, 0 \leq q \leq 3, d = 0$ and $0 \leq P \leq 2, 0 \leq Q \leq 2, D = 1, m = 52$	10091.671	2023412.2883997217

Table 2: SARIMA and Auto ARIMA model parameters and results for The Very Hungry Caterpillar sales data from January 01, 2012 onwards.

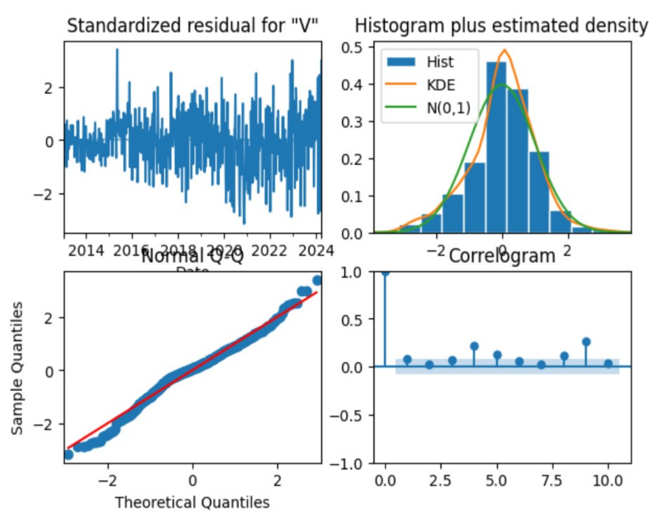


Figure 13: Residuals using SARIMA3 model for The Very Hungry Caterpillar using sales data from January 1, 2012.

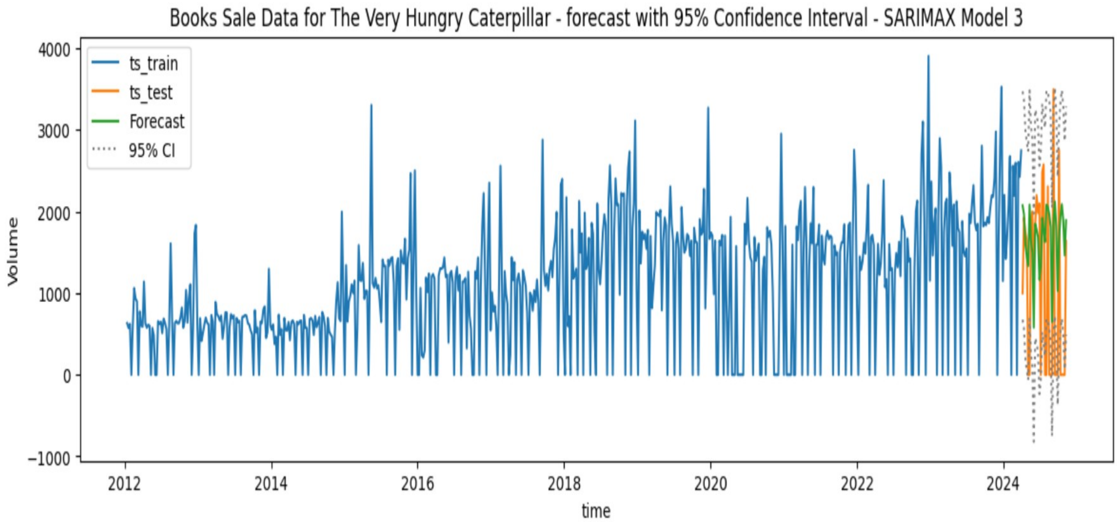


Figure 14: 32-Weeks prediction using SARIMA3 model for The Very Hungry Caterpillar using sales data from January 1, 2012.

3.3.3 Prediction for the Final 32 Weeks

Predictions for the final 32 weeks on the slaes of Book2 are shown in Figure 14. These predictions fit well with test data for this model.

4 Machine Learning & Deep Learning Techniques

For the ML/DL approaches, the training data was required to be upto January 01, 2012. The sales data was resampled on weekly basis and aggregated for sales volume.

4.1 Book 1: The Alchemist

4.1.1 XGBoost

The best hyparameters were optimised for XGBoost method as given below:

colsample_bytree $= 0.6619914885516045$
$\gamma = 4.559754769106771$
max_depth $= 5.0$
min_child_weight $= 0.0$
reg_alpha $= 55.0$

The above parameters produced a model that yielded MAE $= 373.8087834548489$,

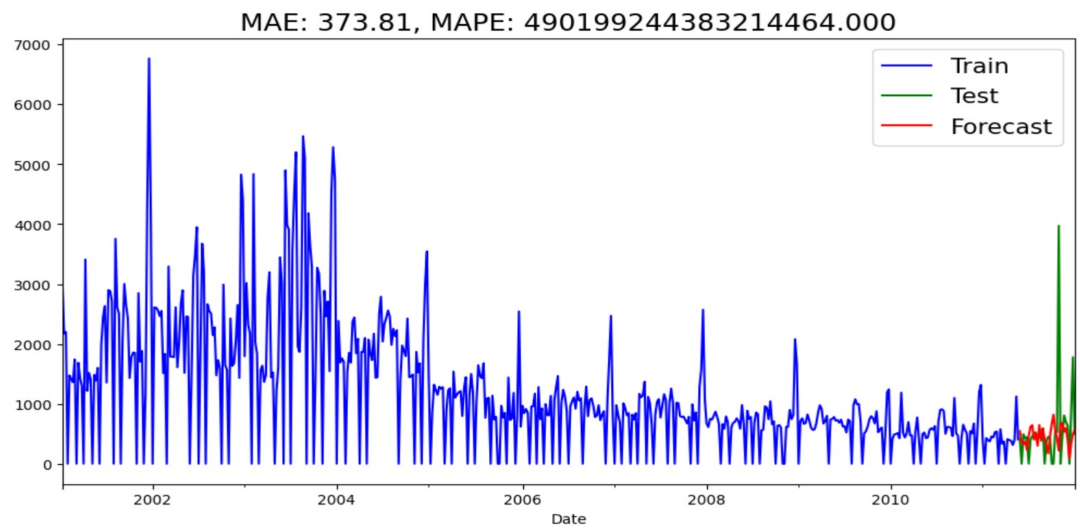


Figure 15: 32-Weeks predictions using XGBoost model, with optimised parameters, for The Alchemist using sales data up to January 1, 2012.

and MAPE = $4.9019924438321446e+17$, with the original data and predictions depicted in Figure 15.

4.1.2 LSTM

The best model that keras tuner provided had parameters shown in Figure 16. This model returned MAE = 0.0511610590852797, and MAPE = 13278501.345. However, we could not plot its predictions.

4.2 Book 2: The Very Hungry Caterpillar

4.2.1 XGBoost

The best hyparameters were optimised for XGBoost method as given below:

colsample_bytree = 0.5143506759311699
$\gamma = 3.676738934958347$
max_depth = 3.0
min_child_weight = 9.0
reg_alpha = 46.0

The above parameters produced a model that yielded MAE = 686.1561167768734, and MAPE = $1.4255637658357724e+18$, with the original data and predictions depicted in Figure 17.

Model: "sequential"

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 12, 84)	28,896
lstm_1 (LSTM)	(None, 12, 108)	83,376
lstm_2 (LSTM)	(None, 12, 100)	83,600
lstm_3 (LSTM)	(None, 12, 68)	45,968
lstm_4 (LSTM)	(None, 84)	51,408
dropout (Dropout)	(None, 84)	0
dense (Dense)	(None, 1)	85

```
 Total params: 293,333 (1.12 MB)
 Trainable params: 293,333 (1.12 MB)
 Non-trainable params: 0 (0.00 B)
None
```

Figure 16: Best model parameters reported by Keras Tuner for The Alchemist using sales data up to January 1, 2012.

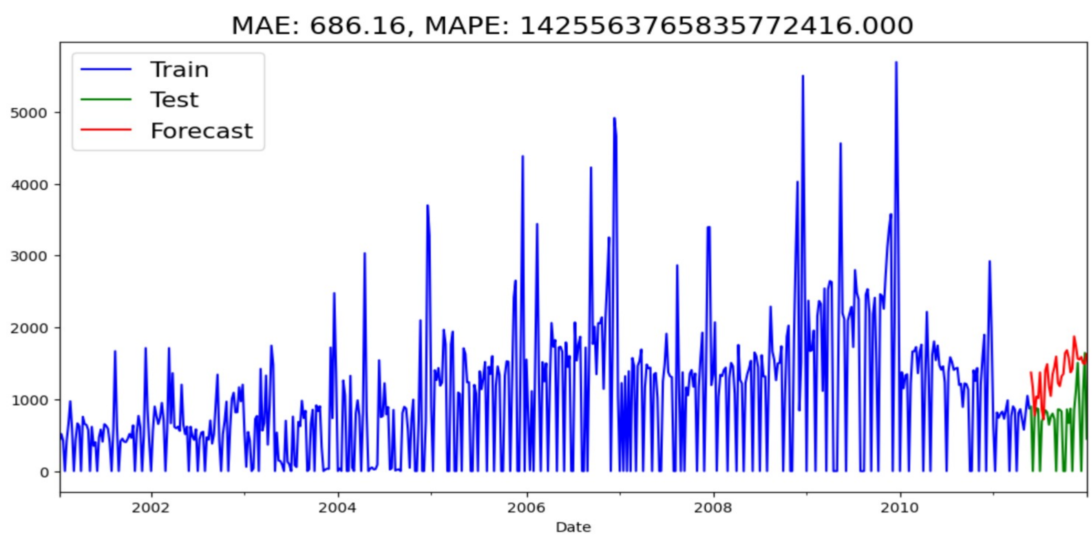


Figure 17: 32-Weeks predictions using XGBoost model, with optimised parameters, for The Very Hungry Caterpillar using sales data up to January 1, 2012.

Model: "sequential"

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 12, 84)	28,896
lstm_1 (LSTM)	(None, 12, 108)	83,376
lstm_2 (LSTM)	(None, 12, 100)	83,600
lstm_3 (LSTM)	(None, 12, 68)	45,968
lstm_4 (LSTM)	(None, 84)	51,408
dropout (Dropout)	(None, 84)	0
dense (Dense)	(None, 1)	85

Total params: 293,333 (1.12 MB)
Trainable params: 293,333 (1.12 MB)
Non-trainable params: 0 (0.00 B)

Figure 18: Best model parameters reported by Keras Tuner for The Very Hungry Caterpillar using sales data up to January 1, 2012.

4.2.2 LSTM

The best model that keras tuner provided had parameters shown in Figure 18. This model returned MAE = 0.048051972761750224, and MAPE = 12795968.35. However, we could not plot its predictions.

5 Hybrid Models

With the hybrid models, it was required to first fit a SARIMA model over the complete sales data for each book and then find the best LSTM model on residuals of the SARIMA model in a sequential approach.

5.1 The Sequential Approach

5.1.1 Book 1: The Alchemist

The weekly sales data for Book1 is depicted in Figure 19.

1. **The SARIMA Model**

   Seasonal decomposition was obtained (in Figure 20) using the additive model as the plot of weekly sales suggests this. We searched the SARIMA model through Auto ARIMA, which provided the model ARIMA(4, 1, 3), (2, 0, 0)[52] (Figure 21). The residuals of the above ARIMA model are plotted in Figure 22.

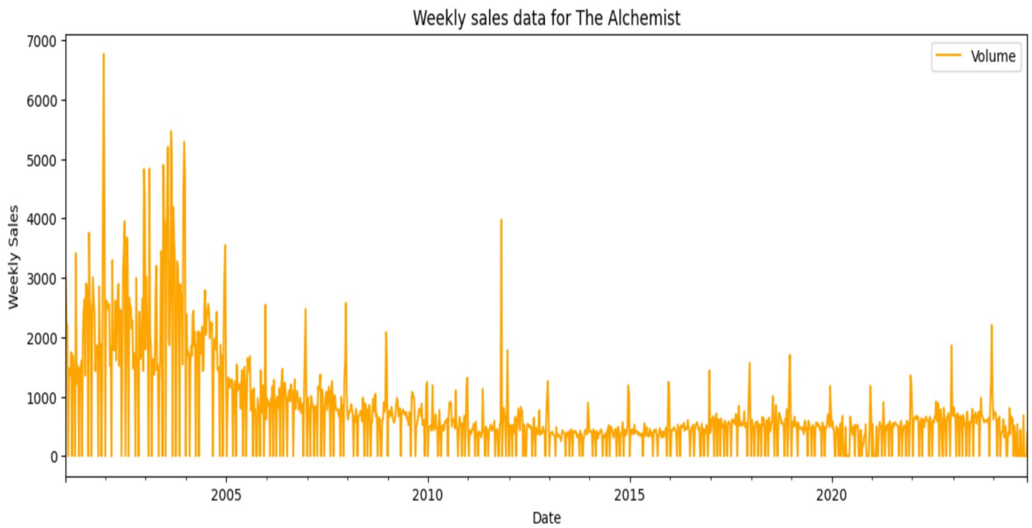


Figure 19: Weekly sales data for The Alchemist for the whole duration.

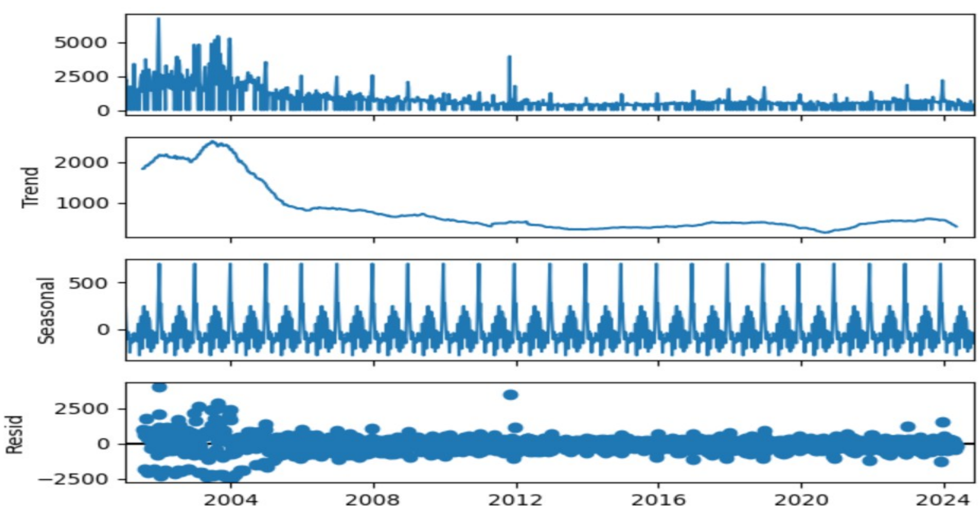


Figure 20: Residuals of the ARIMA model for weekly sales data of The Alchemist for the whole duration.

SARIMAX Results

Dep. Variable:	Volume	**No. Observations:**	1212
Model:	SARIMAX(4, 1, 3)x(2, 0, [], 52)	**Log Likelihood**	-9333.458
Date:	Tue, 04 Feb 2025	**AIC**	18688.917
Time:	18:20:51	**BIC**	18745.008
Sample:	01-13-2001 - 03-30-2024	**HQIC**	18710.036
Covariance Type:	opg		

	coef	std err	z	P>|z|	[0.025	0.975]
intercept	-0.5098	5.335	-0.096	0.924	-10.966	9.947
ar.L1	-1.6539	0.029	-57.092	0.000	-1.711	-1.597
ar.L2	-0.8924	0.041	-22.004	0.000	-0.972	-0.813
ar.L3	-0.4278	0.033	-12.785	0.000	-0.493	-0.362
ar.L4	-0.2956	0.017	-17.369	0.000	-0.329	-0.262
ma.L1	0.7847	0.031	25.358	0.000	0.724	0.845
ma.L2	-0.7222	0.019	-38.164	0.000	-0.759	-0.685
ma.L3	-0.6674	0.030	-21.937	0.000	-0.727	-0.608
ar.S.L52	0.2533	0.015	17.382	0.000	0.225	0.282
ar.S.L104	0.2771	0.020	13.622	0.000	0.237	0.317
sigma2	2.863e+05	5175.219	55.331	0.000	2.76e+05	2.96e+05

Ljung-Box (L1) (Q):	0.07	**Jarque-Bera (JB):**	5362.38
Prob(Q):	0.80	**Prob(JB):**	0.00
Heteroskedasticity (H):	0.10	**Skew:**	0.42
Prob(H) (two-sided):	0.00	**Kurtosis:**	13.27

Figure 21: The SARIMA model searched through Auto ARIMA for weekly sales data of The Alchemist for the whole duration.

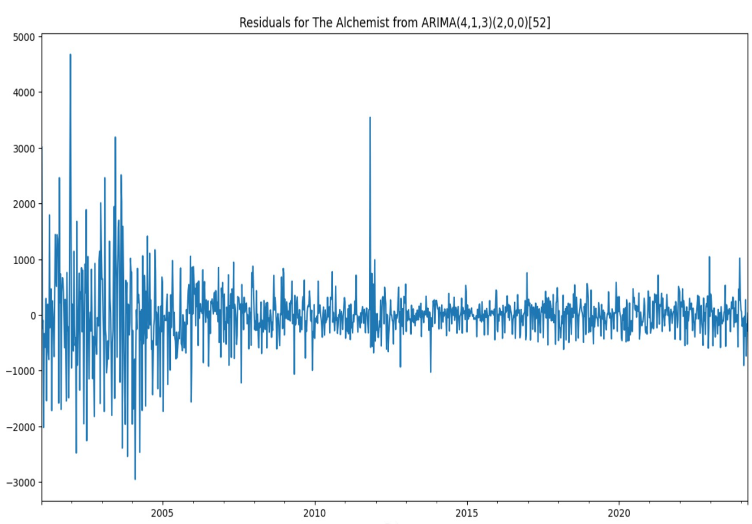


Figure 22: Residuals from SARIMA model in sequential approach for The Alchemist for the whole duration.

Model: "sequential"

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 12, 4)	96
lstm_1 (LSTM)	(None, 12, 100)	42,000
lstm_2 (LSTM)	(None, 12, 4)	1,680
lstm_3 (LSTM)	(None, 12, 4)	144
lstm_4 (LSTM)	(None, 12, 4)	144
lstm_5 (LSTM)	(None, 68)	19,856
dropout (Dropout)	(None, 68)	0
dense (Dense)	(None, 1)	69

```
Total params: 63,989 (249.96 KB)
Trainable params: 63,989 (249.96 KB)
Non-trainable params: 0 (0.00 B)
None
```

Figure 23: Best LSTM model of weekly sales data searched by Keras tuner for The Alchemist for the whole duration.

2. **The LSTM Model**

   The Keras tuner provided the best model with parameter listed in Figure 23.

5.1.2 Book 2: The Very Hungry Caterpillar

1. **The SARIMA Model**

   The sales data for Book2 is given in Figure 24, with the seasonal decomposition in Figure 25. The SARIMA model, suggested by Auto ARIMA, with parameters ARIMA(2, 1, 4), (2, 0, 0)[52], generated residuals which are plotted in Figure 26.

2. **The LSTM Model**

   The best model searched by Keras tuner for Book2 is detailed in Figure 27.

5.1.3 A Note on the Sequential Approach in Hybrid Modelling

We could not complete the sequential approach of hybrid modelling for the two books due to problems with Internet. We would like to improve this work by plotting results of the LTSM models for the residuals of the ARIMA models

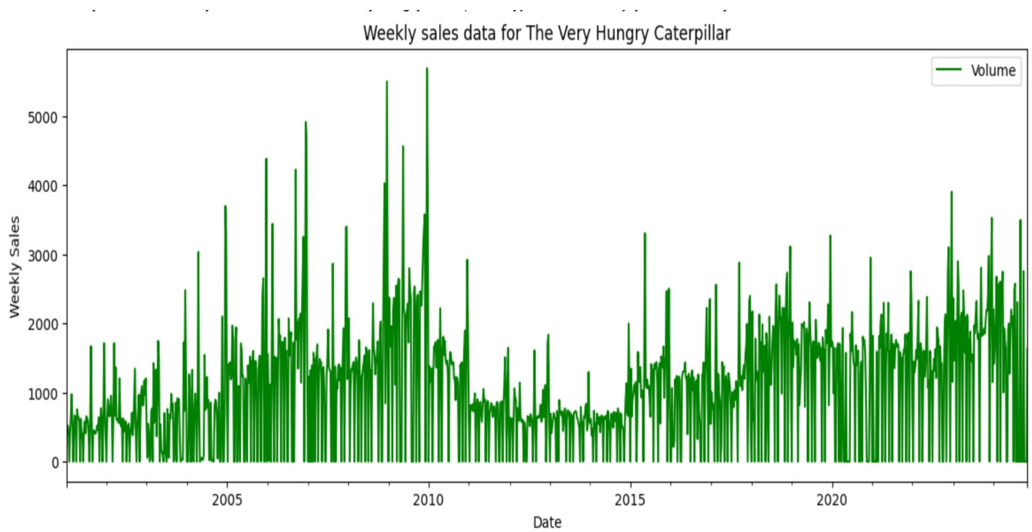


Figure 24: Weekly sales data for The Very Hungry Caterpillar for the whole duration.

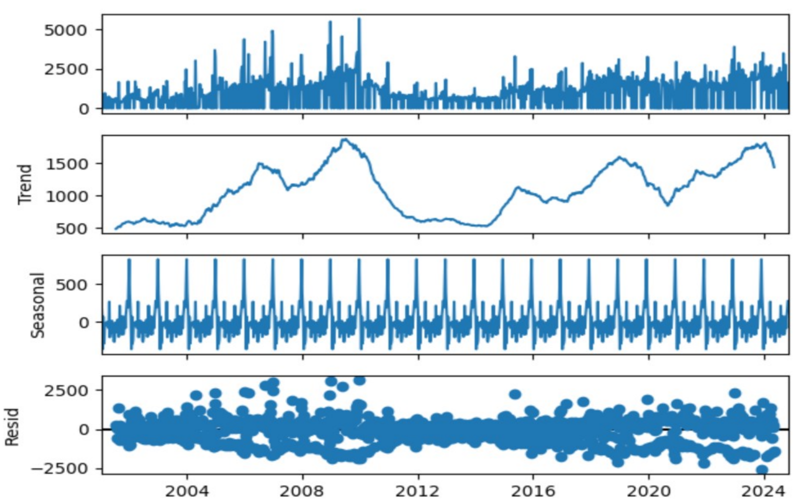


Figure 25: Residuals of the ARIMA model for weekly sales data of The Very Hungry Caterpillar for the whole duration.

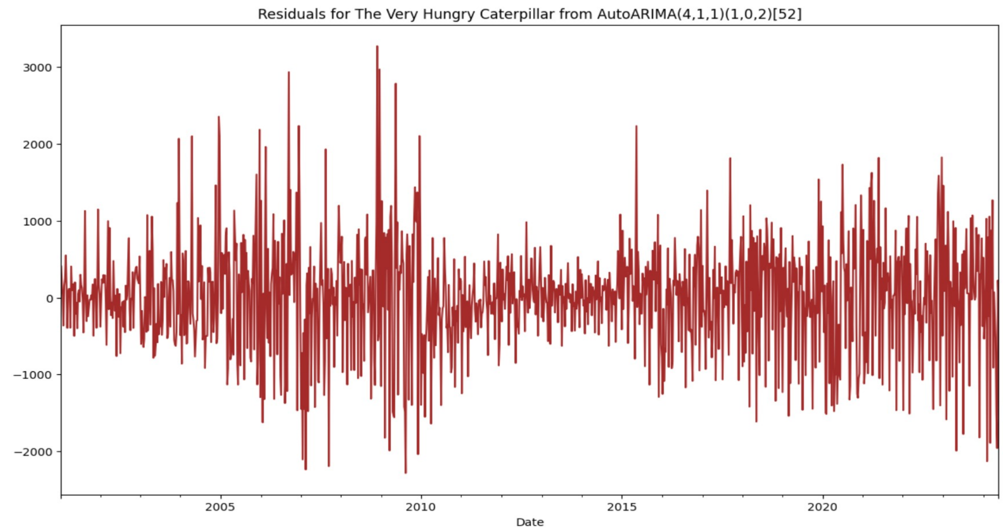


Figure 26: Residuals from SARIMA model in sequential approach for The Very Hungry Caterpillar for the whole duration.

Model: "sequential"

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 12, 4)	96
lstm_1 (LSTM)	(None, 12, 100)	42,000
lstm_2 (LSTM)	(None, 12, 4)	1,680
lstm_3 (LSTM)	(None, 12, 4)	144
lstm_4 (LSTM)	(None, 12, 4)	144
lstm_5 (LSTM)	(None, 68)	19,856
dropout (Dropout)	(None, 68)	0
dense (Dense)	(None, 1)	69

Total params: 63,989 (249.96 KB)
Trainable params: 63,989 (249.96 KB)
Non-trainable params: 0 (0.00 B)
None

Figure 27: Best LSTM model of weekly sales data searched by Keras tuner for The Very Hungry Caterpillar for the whole duration.

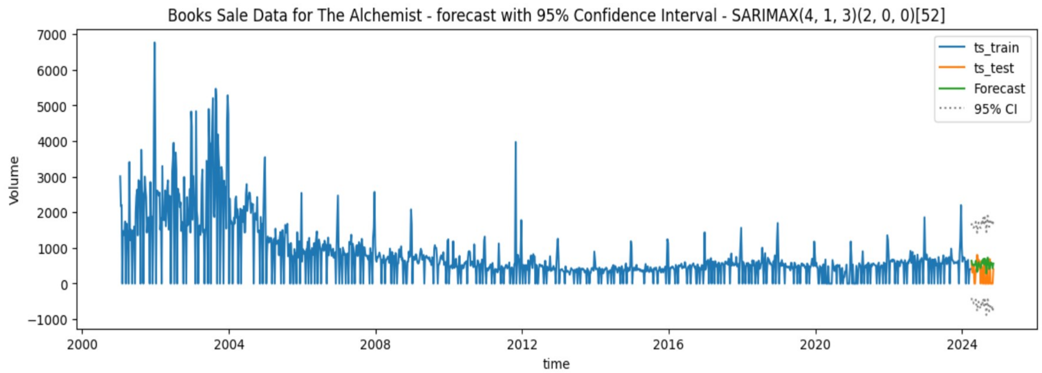


Figure 28: Forecast plot of SARIMA best model parameters for The Alchemist complete time series sales data.

suggested for the two books. This will assist in proper comparison of this approach with other approaches.

5.2 The Parallel Approach

5.2.1 Book 1: The Alchemist

1. **The SARIMA Model**

   The best SARIMA model that we had found in Section 1 produced performance parameters, as given in Figure 21, and its forecast is depicted in Figure 28. This model produced MAE = 338.0684798244364, and MAPE = inf. The MAPE being infinite could be expected due to the resampling of data with zero fillings.

2. **The LSTM Model**

   The Keras tuner returned the best model with parameters in Figure 29: MAE = 0.04871254863217473, and MAPE = 18601274.23.

5.2.2 Book 2: The Very Hungry Caterpillar

1. **The SARIMA Model**

   The best SARIMA model that we had found in Section 1 produced performance parameters, as given in Figure 31, and its forecast is depicted in Figure 32. This model produced MAE = 1159.6351258283903, and MAPE = inf.

```
Model: "sequential"
```

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 12, 84)	28,896
lstm_1 (LSTM)	(None, 12, 20)	8,400
lstm_2 (LSTM)	(None, 52)	15,184
dropout (Dropout)	(None, 52)	0
dense (Dense)	(None, 1)	53

```
 Total params: 52,533 (205.21 KB)
 Trainable params: 52,533 (205.21 KB)
 Non-trainable params: 0 (0.00 B)
None
```

Figure 29: LSTM best model parameters for The Alchemist complete time series sales data.

2. **The LSTM Model**

   The Keras tuner returned the best model with parameters in Figure 30: MAE = 0.09697699137032031, and MAPE = 188710.825859375.

6 Monthly Prediction

The monthly sales data for books was obtained by first resampling the whole data from all worksheets on a monthly basis and then by aggregating the volume of sales. Each of the two books had 287 monthly records which is about 24 years.

6.1 Book 1: The Alchemist

Table 3 lists the parameters of the best XGBoost model and the best SARIMA model, searched through Auto ARIMA. The XGBoost Model brought better forecast for future 8 months (Figure 33) than the SARMAX model (Figure 34).

6.2 Book 2: The Very Hungry Caterpillar

For Book2, parameters of both models are given in Table 4. The forecasts for XGBoost and SARIMA models are depicted in Figures 35 and 36 respectively. Again, the XGBoost model produced a better 8-months forecast with relatively better MAE and MAPE scores.

Model: "sequential"

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 12, 84)	28,896
lstm_1 (LSTM)	(None, 12, 20)	8,400
lstm_2 (LSTM)	(None, 52)	15,184
dropout (Dropout)	(None, 52)	0
dense (Dense)	(None, 1)	53

Total params: 52,533 (205.21 KB)
Trainable params: 52,533 (205.21 KB)
Non-trainable params: 0 (0.00 B)
None

Figure 30: LSTM best model parameters for The Very Hungry Caterpillar complete time series sales data.

SARIMAX Results

Dep. Variable:	Volume	**No. Observations:**	1212
Model:	SARIMAX(2, 1, 4)x(2, 0, [], 52)	**Log Likelihood**	-9625.995
Date:	Tue, 04 Feb 2025	**AIC**	19271.991
Time:	23:30:43	**BIC**	19322.983
Sample:	01-13-2001	**HQIC**	19291.191
	- 03-30-2024		
Covariance Type:	opg		

	coef	**std err**	**z**	**P>|z|**	**[0.025**	**0.975]**
intercept	1.2764	2.963	0.431	0.667	-4.531	7.083
ar.L1	0.2506	0.001	214.497	0.000	0.248	0.253
ar.L2	-0.9998	0.001	-1550.850	0.000	-1.001	-0.999
ma.L1	-1.1008	0.031	-35.859	0.000	-1.161	-1.041
ma.L2	1.1254	0.039	28.949	0.000	1.049	1.202
ma.L3	-0.8293	0.039	-21.309	0.000	-0.906	-0.753
ma.L4	-0.0798	0.031	-2.608	0.009	-0.140	-0.020
ar.S.L52	0.1669	0.028	5.960	0.000	0.112	0.222
ar.S.L104	0.1200	0.033	3.665	0.000	0.056	0.184
sigma2	5.73e+05	5.38e-05	1.07e+10	0.000	5.73e+05	5.73e+05

Ljung-Box (L1) (Q):	0.03	**Jarque-Bera (JB):**	169.99
Prob(Q):	0.85	**Prob(JB):**	0.00
Heteroskedasticity (H):	1.13	**Skew:**	0.25
Prob(H) (two-sided):	0.21	**Kurtosis:**	4.77

Figure 31: SARIMA best model parameters for The Very Hungry Caterpillar complete time series sales data.

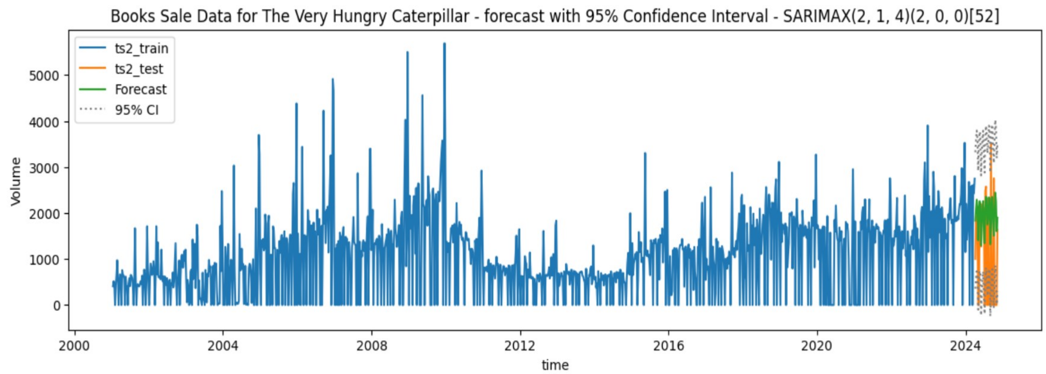


Figure 32: Forecast plot of SARIMA best model parameters for The Very Hungry Caterpillar complete time series sales data.

Model	Model Parameters	Model Performance
XGBoost Model	colsample_bytree = 0.7043466125281708, γ = 1.919205910489704, max_depth = 4.0, min_child_weight = 4.0 and reg_alpha = 48	MAE = 1391.8590767578526, $MAPE$ = 2.1459010336556315
SARIMA Model	$(0, 1, 1)(0, 0, 0)[52]$	AIC = 5008.228, MAE = 1789.4889416645667, $MAPE$ = 287.8809431725168

Table 3: Models and performance results for The Alchemist monthly sales data using the XGBoost and SARMIA models.

Model	Model Parameters	Model Performance
XGBoost Model	colsample_bytree = 0.9158503464804676, γ = 3.804829783484248, max_depth = 6.0, min_child_weight = 4.0 and reg_alpha = 177	MAE = 3913.7522160925632, $MAPE$ = 1.2869840099536576
SARIMA Model	$(3, 0, 2)(0, 0, 0)[52]$	AIC = 5128.673, MAE = 6576.567882044389, $MAPE$ = 982.4620377332518

Table 4: Models and performance results for The Very Hungry Caterpillar monthly sales data using the XGBoost and SARMIA models.

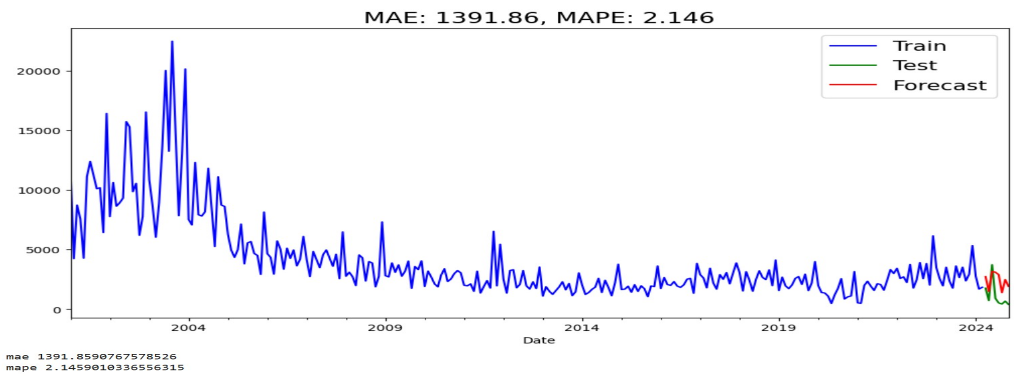


Figure 33: 8-month forecast by the XGBoost model for The Alchemist complete time series sales data.

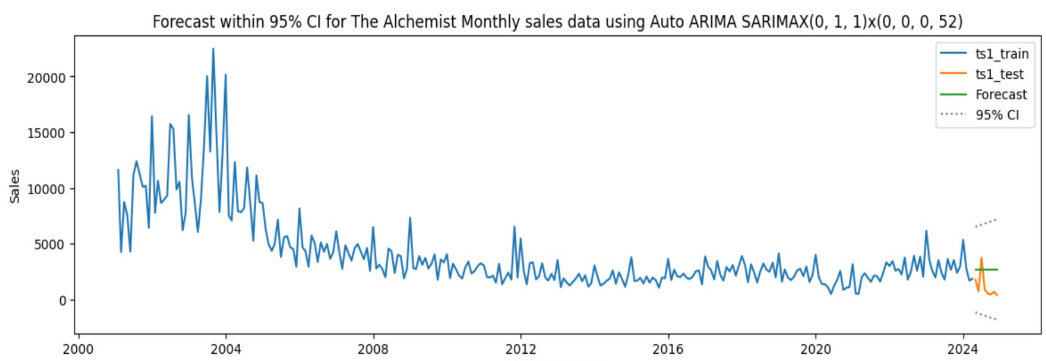


Figure 34: 8-month forecast by the SARIMA model for The Alchemist complete time series sales data.

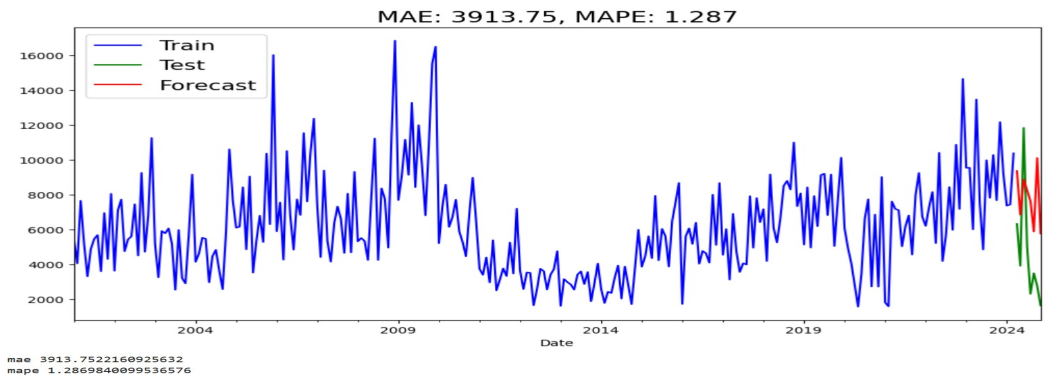


Figure 35: 8-month forecast by the XGBoost model for The Very Hungry Caterpillar complete time series sales data.

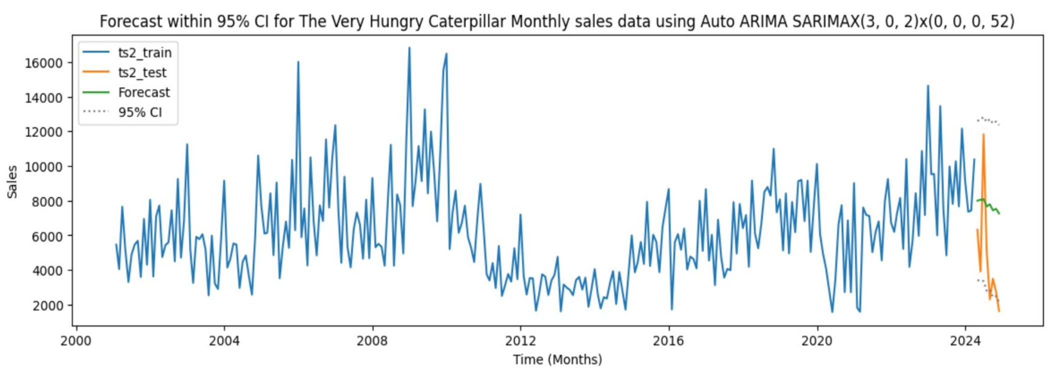


Figure 36: 8-month forecast by the SARIMA model for The Very Hungry Caterpillar complete time series sales data.

7 Conclusions

For the weekly sales data, the classical approaches of SARIMA model provided a better forecast for the last 32 weeks which agreed quite well with the test data. However, this conclusion is limited for the sales data starting from January 1, 2012 onwards. For monthly predictions of the last 8- months, XGBoost performed better than the best SARIMA for both books over the whole sales data for 24 years. This project was a good exercise in time series analysus and forecasting, but we would like to alter our analysis once we complete the forecast using the hybrid approaches for the two books.